Notes

Recap of Previous Class Topics

Overview of Previously Covered Concepts

In the previous session, we covered several fundamental topics in electrostatics. These included:

- **Properties of Charges:** We discussed the basic characteristics of electric charges, such as their additive nature, conservation, and quantization.
- **Methods of Charging:** We explored different ways to charge an object, including charging by friction, conduction, and induction.
- **Conductors and Insulators:** A brief overview of materials based on their ability to conduct electricity was provided, distinguishing between conductors and insulators.
- **Coulomb's Law:** We introduced Coulomb's law, which quantifies the force between two point charges.
  - The magnitude of the electrostatic force (F) between two point charges (q_1 and q_2) separated by a distance (r) is given by: $F = k\frac{|q_1 q_2|}{r^2}$ where k is Coulomb's constant ($k = \frac{1}{4\pi\epsilon_0} \approx 9 \times 10^9$ N m^2/C^2).
- **Coulomb's Law in Vector Form:** We also discussed how to express Coulomb's law in vector form, which accounts for both the magnitude and direction of the force.
- **Questions Based on Coulomb's Law:** We practiced various types of problems involving Coulomb's law, including scenarios of equilibrium when multiple charges are present.

Homework Discussion: Force and Displacement of Charges

Introduction

This section discusses a homework problem related to charges placed on a line and subjected to horizontal displacement. The core of the problem is to determine the relationship between the force experienced by a charge and its displacement.

Problem Statement

Consider a scenario where charges are placed on the same line. A horizontal displacement is given to one of these charges. The task is to determine the relationship between the force (F) acting on the displaced charge and its displacement (X).

Specifically, for two different cases (left-hand side and a second case), we need to identify whether:

- F is proportional to X ($F \propto X$)
- F is proportional to -X ($F \propto -X$)

Key Question

For both scenarios, what is the correct statement regarding the proportionality of force to displacement?

Importance of Consistent Effort

The instructor emphasized that consistent, sincere effort on small daily tasks, like solving homework problems, is crucial for achieving excellent results in exams like JEE. There are no secret formulas or shortcuts; success comes from diligent daily practice.

Introduction to Electric Field

Understanding Fields through Analogies

To understand the concept of an electric field, we can consider several analogies:

- **Wi-Fi Router:** A Wi-Fi router creates a field around it. Within this field, devices can connect to the internet. The strength of the connection (and thus the internet speed) is strongest near the router and decreases as you move further away. Eventually, outside a certain range, connection is lost.
- **Fragrance (Perfume/Coffee Shop):** A perfume or coffee shop emits a fragrance that can be detected within a certain zone around it. The intensity of the smell is stronger closer to the source and fades with distance. Beyond a certain point, the fragrance is undetectable.
- **Gravitational Field:** Planets create a gravitational field around them. Within this field, objects experience the planet's gravitational pull. The strength of this pull varies with distance from the planet.
- **Magnetic Field:** Magnets create a magnetic field around them, where magnetic forces can be experienced.

Definition of Electric Field

Similar to these examples, a **charge** creates an **electric field** around it. This electric field is a region, zone, or area in space where another charge would experience an electrostatic force. This force can be either attractive or repulsive, depending on the nature of the charges involved.

- **Key Characteristic:** If you are within the electric field of a charge, you will experience an electrostatic force. If you are outside this field, you will not experience any electrostatic force from that charge.
- **Strength Variation:** The strength of the electric field, and consequently the electrostatic force experienced, is not uniform throughout the field. It is generally stronger closer to the source

Electric Field Intensity

Definition and Measurement

To measure the strength of an electric field, we use a quantity called **electric field intensity**, often simply referred to as the **electric field** (E).

To determine the electric field intensity at a specific location (e.g., point A) within an electric field, a small positive charge, known as a **test charge** (Q_0), is placed at that location.

- **Test Charge (Q_0):** A very small positive charge. It is kept small to ensure it does not significantly disturb the existing electric field being measured.
- **Force on Test Charge:** When placed in an electric field, the test charge (Q_0) experiences a force (F).

Formula for Electric Field Intensity

The electric field intensity (E) is defined as the force experienced per unit test charge:

$$E = \frac{F}{Q_0}$$

Where:

- E is the electric field intensity.
- F is the force experienced by the test charge.
- Q_0 is the magnitude of the test charge.

Characteristics of Electric Field Intensity

- **Vector Quantity:** Electric field intensity is a vector quantity, meaning it has both magnitude and direction.
- **SI Units:** The SI units for electric field intensity are:
  - Newton per Coulomb (N/C)
  - Volt per meter (V/m)

Force on a Charge in an Electric Field

If a charge (Q) is placed in an electric field (E), it will experience a force (F). This force can be calculated by rearranging the electric field intensity formula:

$$F = Q \cdot E$$

Direction of Force on a Charge

- **Positive Charge:** If the charge (Q) is positive, the force (F) will be in the **same direction** as the electric field (E).
  - Example: If the electric field is towards the right, a positive charge will experience a force towards the right.
- **Negative Charge:** If the charge (Q) is negative, the force (F) will be in the **opposite direction** to the electric field (E).
  - Example: If the electric field is towards the right, a negative charge will experience a force towards the left.

This principle is fundamental and will be applied repeatedly in electromagnetism.

Problem: Sphere in Electric Field

Problem Statement

A small sphere of mass 1 gram carries a charge of +6 microcoulombs. The sphere is suspended by a string in an electric field of 400 N/C, acting downward. We need to find the tension in the string.

Given Values

- Mass of the sphere (m) = 1 gram = 1×10^{-3} kg
- Charge of the sphere (q) = +6 microcoulombs = $+6 \times 10^{-6}$ C
- Electric field (E) = 400 N/C (downward)
- Acceleration due to gravity (g) = 10 m/s^2

Concept: Equilibrium of Forces

Since the sphere is suspended and at rest, it is in a state of equilibrium. This means the net force acting on the sphere is zero. According to Newton's first law, the sum of all forces in both the horizontal and vertical directions must be zero.

Free Body Diagram (FBD) Analysis

We will analyze the forces acting on the sphere:

1. **Gravitational Force (Weight):** This force acts vertically downward and is given by $W = mg$.
2. **Tension Force:** This force acts along the string, vertically upward, away from the sphere. Let's denote it as T.
3. **Electric Force:** Since the sphere has a positive charge ($q > 0$) and the electric field (E) is directed downward, the electric force (F_e) will also act in the direction of the electric field, i.e., vertically downward. The magnitude of this force is given by $F_e = qE$.

forces.

Upward forces: T Downward forces: $W + F_e = mg + qE$

Therefore, for equilibrium: $T = mg + qE$

Calculation of Tension

Now, we substitute the given values into the equation: $T = (1 \times 10^{-3}\ \text{kg})(10\ \text{m/s}^2) + (6 \times 10^{-6}\ \text{C})(400\ \text{N/C})$

Calculate the gravitational force: $mg = 1 \times 10^{-3} \times 10 = 10 \times 10^{-3} = 1 \times 10^{-2}$ N

Calculate the electric force: $qE = 6 \times 10^{-6} \times 400 = 2400 \times 10^{-6} = 2.4 \times 10^{3} \times 10^{-6} = 2.4 \times 10^{-3}$ N

Now, sum the forces to find the tension: $T = 1 \times 10^{-2}\ \text{N} + 2.4 \times 10^{-3}\ \text{N}$

To add these values, we can express them with the same power of 10: $1 \times 10^{-2} = 10 \times 10^{-3}$

So, $T = 10 \times 10^{-3}\ \text{N} + 2.4 \times 10^{-3}\ \text{N}$ $T = (10 + 2.4) \times 10^{-3}\ \text{N}$ $T = 12.4 \times 10^{-3}\ \text{N}$

Alternatively, we can express it as: $T = 0.01\ \text{N} + 0.0024\ \text{N}$ $T = 0.0124\ \text{N}$

This can also be written as: $T = 1.24 \times 10^{-2}\ \text{N}$

Final Answer

The tension in the string is 1.24×10^{-2} N.

Electric Field Due to a Point Charge

Definition and Formula

The electric field produced by a point charge Q fixed at a location is a fundamental concept. This charge acts as the source of the electric field. To determine the electric field intensity at a point A, located at a distance R from the charge Q, we use a test charge q_0.

According to Coulomb's Law, the force (F) experienced by the test charge q_0 due to the source charge Q is given by: $F = \frac{kQq_0}{R^2}$ where k is Coulomb's constant, $k = \frac{1}{4\pi\epsilon_0}$.

The electric field (E) at point A is defined as the force per unit test charge: $E = \frac{F}{q_0}$ Substituting the expression for F: $E = \frac{\frac{kQq_0}{R^2}}{q_0}$ The test charge q_0 cancels out, giving the formula for the electric field produced by a point charge: $E = \frac{kQ}{R^2}$ This formula is crucial and will be used extensively in various chapters.

- **For a positive charge ($Q > 0$):** The electric field is **radially outward** from the charge. This can be understood by placing a positive test charge q_0 near a positive source charge; the repulsive force on q_0 will be directed away from Q.
- **For a negative charge ($Q < 0$):** The electric field is **radially inward** towards the charge. If a positive test charge q_0 is placed near a negative source charge, the attractive force on q_0 will be directed towards Q.

In summary:

- Positive charge: Electric field lines point away from the charge.
- Negative charge: Electric field lines point towards the charge.

Calculating Net Electric Field at Point C

Problem Statement

We have charges at point A and point B on a line. We need to find the net electric field at point C.

- Charge at A: $-18\mu C$
- Charge at B: $9\mu C$
- Distance between A and B: 3 meters
- Distance between B and C: 3 meters

Superposition Principle

The net electric field at point C will be the vector sum of the electric field produced by the charge at A (E_A) and the electric field produced by the charge at B (E_B).

$E_{net} = E_A + E_B$

Direction of Electric Fields

1. **Electric Field due to Charge at A (E_A):**
   - The charge at A is negative ($-18\mu C$).
   - For a negative charge, the electric field lines point *towards* the charge.
   - Therefore, E_A will be directed towards A (left-hand side).
2. **Electric Field due to Charge at B (E_B):**
   - The charge at B is positive ($9\mu C$).
   - For a positive charge, the electric field lines point *away* from the charge.

To perform the vector sum, we adopt a sign convention:

- Right-hand side: Positive
- Left-hand side: Negative

So, E_B will be positive, and E_A will be negative.

Calculation of Electric Fields

The formula for the electric field due to a point charge is:

$E = \frac{kQ}{r^2}$

where:

- k is Coulomb's constant (9×10^9 Nm^2/C^2)
- Q is the magnitude of the charge
- r is the distance from the charge to the point where the electric field is being calculated.

1. **Electric Field due to Charge at B (E_B):**
   - Charge $Q_B = 9\mu C = 9 \times 10^{-6}$ C
   - Distance r_B = 3 meters
   - Direction: Right (positive)

   $E_B = \frac{k \times (9 \times 10^{-6})}{3^2} = \frac{k \times 9 \times 10^{-6}}{9} = k \times 10^{-6}$ N/C

2. **Electric Field due to Charge at A (E_A):**
   - Charge $Q_A = -18\mu C$ (we use the magnitude for calculation, direction is handled by sign convention)
   - Distance r_A = Distance(A to B) + Distance(B to C) = 3 + 3 = 6 meters
   - Direction: Left (negative)

   $E_A = -\frac{k \times (18 \times 10^{-6})}{6^2} = -\frac{k \times 18 \times 10^{-6}}{36} = -\frac{k \times 10^{-6}}{2}$ N/C

Net Electric Field (E_{net})

$E_{net} = E_B + E_A$ $E_{net} = (k \times 10^{-6}) + (-\frac{k \times 10^{-6}}{2})$ $E_{net} = k \times 10^{-6}(1 - \frac{1}{2})$ $E_{net} = k \times 10^{-6} \times \frac{1}{2}$

Substitute the value of $k = 9 \times 10^9$ Nm^2/C^2:

$E_{net} = (9 \times 10^9) \times 10^{-6} \times \frac{1}{2}$ $E_{net} = 9 \times 10^3 \times \frac{1}{2}$ $E_{net} = 4.5 \times 10^3$ N/C

net electric field is towards the right-hand side.

Electric Field at the Center of a Hexagon with Charges

Introduction

When calculating the net electric field, it's crucial to consider the direction of the electric field produced by individual charges and then perform a vector sum. For a negative charge, the electric field points towards the charge. For a positive charge, the electric field points away from the charge.

Problem Statement

Consider a regular hexagon with charges placed at its corners. We need to determine if the electric field at the center of the hexagon is zero.

Analysis of Electric Fields

Let's label the corners of the hexagon A, B, C, D, E, and F. The electric field produced by a positive charge is directed away from it. The formula for the electric field due to a point charge is given by: $E = \frac{kq}{r^2}$ where:

- k is Coulomb's constant ($k = \frac{1}{4\pi\epsilon_0}$)
- q is the magnitude of the charge
- r is the distance from the charge to the point where the electric field is being calculated.

In a regular hexagon, all corners are equidistant from the center. If the charges at points B, D, E, and A are identical, their electric fields at the center will have the same magnitude. However, their directions will be such that they cancel each other out. For example, the electric field from B will cancel the electric field from E, and the electric field from A will cancel the electric field from D.

Resultant Electric Field

If charges at points A, B, D, and E cancel each other out, the net electric field at the center will be solely due to the remaining charge(s). If only one charge remains, say at point C, then the net electric field will be the electric field produced by the charge at C.

Let the charge at C be q and the distance from the center to any corner be a. Then the net electric field (E_{net}) at the center will be: $E_{net} = E_C = \frac{kq}{a^2}$ Substituting $k = \frac{1}{4\pi\epsilon_0}$: $E_{net} = \frac{1}{4\pi\epsilon_0}\frac{q}{a^2}$ If the charge at C is positive, the direction of the electric field will be downwards (e.g., in the $-j$ direction).

Example Problem: Five Charges on a Hexagon

Notes

Solution Approach

Similar to the previous case, we can use the principle of superposition. The distance from the center of a regular hexagon to any of its vertices is equal to its side length. So, in this case, $r = 2a$.

The electric field due to each charge at the center will have the same magnitude: $E = \frac{k(+q)}{(2a)^2} = \frac{kq}{4a^2}$ If we consider the six vertices, and one vertex is empty (or has no charge), the electric fields from the charges at opposite vertices will cancel each other out, except for the one that doesn't have an opposite counterpart with a charge. For example, if the charge at one vertex is missing, the electric field that would have been produced by a charge at that missing vertex (if it were present) would be the resultant electric field, but in the opposite direction.

Let's assume the missing charge is at the top vertex. Then the electric fields from the other five charges will sum up such that the net electric field is equivalent to the electric field produced by a single charge $+q$ placed at the vertex *opposite* to the missing vertex, but pointing in the direction *away* from that opposite vertex.

So, the net electric field at the center will be: $E_{net} = \frac{k(+q)}{(2a)^2} = \frac{1}{4\pi\epsilon_0}\frac{q}{4a^2}$ $E_{net} = \frac{q}{16\pi\epsilon_0 a^2}$ This corresponds to option B in a multiple-choice question.

Property of Regular Hexagons

A regular hexagon can be divided into six equilateral triangles, where the side length of each triangle is equal to the side length of the hexagon. This property ensures that all vertices are equidistant from the center, simplifying electric field calculations due to symmetry.

Non-Regular Hexagons

If the hexagon is not regular, or if the charges are not symmetrically placed, then the individual electric fields from each charge must be calculated, and their vector sum must be performed. This would require more geometric data and would be a more lengthy calculation, as each component would need to be resolved and summed individually.

Electric Field Due to Point Charges - Homework Problem

Problem Statement

Consider the Earth's globe with radius R. Multiple point charges, each of magnitude Q, are placed along the equatorial plane. These charges are separated by an angular distance of 30°, covering the entire equatorial plane.

Part 1: Electric Field at the Center of the Globe

placed on the equatorial plane, there is an identical charge Q diametrically opposite to it. The electric fields produced by these pairs of charges at the center will be equal in magnitude and opposite in direction, thus canceling each other out. This symmetry holds true for all charges placed uniformly around the equator.

Part 2: Electric Field at Any One Pole (Homework Question)

What is the electric field at any one of the poles of the globe?

- **Hint:** This is a more complex problem requiring vector addition of electric fields from multiple charges. Consider the geometry and the contribution of each charge to the electric field at the pole. This question is assigned as homework and will be discussed in the next class.

Continuous Charge Distribution

Introduction

We have previously discussed point charges and multiple point charges. Now, we will move towards understanding continuous charge distribution, where charge is spread uniformly over a length, area, or volume. There are three main types of continuous charge distributions:

1. **Linear Charge Distribution**
2. **Surface Charge Distribution**
3. **Volume Charge Distribution**

1. Linear Charge Distribution

- **Applicability:** This type of distribution applies to one-dimensional objects such as rods, wires, or thin rings.
- **Concept:** Imagine a rod of total length L with a total charge Q distributed uniformly along its entire length.
- **Linear Charge Density (**
  $lambda$**):** It is defined as the charge per unit length.
  - **Formula:**
    $lambda =$
    $fracQL$
  - **Units:** Coulomb per meter (C/m)
  - **Meaning:** It quantifies how densely the charge is packed per unit length, i.e., how much charge is present for every unit length of the object.

2. Surface Charge Distribution

- **Applicability:** This applies to two-dimensional objects like plates or sheets.

- **Surface Charge Density ($sigma$):** It is defined as the charge per unit area.
  - **Formula:** $sigma = fracQA$
  - **Units:** Coulomb per square meter (C/m^2)
  - **Meaning:** It indicates the amount of charge distributed per unit area of the surface.

3. Volume Charge Distribution

- **Applicability:** This applies to three-dimensional objects where charge is distributed throughout their volume.
- **Concept:** Imagine a solid plastic ball (not hollow) where charge Q is distributed uniformly throughout its entire volume V, not just on its surface.
- **Volume Charge Density (ρ):** It is defined as the charge per unit volume.
  - **Formula:** $rho = fracQV$
  - **Units:** Coulomb per cubic meter (C/m^3)
  - **Meaning:** It represents the amount of charge contained within every unit volume of the object.

Summary of Charge Densities

- **Linear Charge Density ($lambda$):** Charge per unit length (C/m), for linear objects.
- **Surface Charge Density ($sigma$):** Charge per unit area (C/m^2), for plates or surfaces.
- **Volume Charge Density (ρ):** Charge per unit volume (C/m^3), for 3D objects where charge is distributed throughout the volume.

Introduction to Continuous Charge Densities

Understanding Continuous Charge Distributions

When dealing with electric fields, we often encounter situations where charge is not concentrated at a single point but is distributed continuously over a length, area, or volume. To analyze such systems, we use the concept of charge densities:

- **Linear Charge Density ($lambda$):** Charge per unit length. Used for charges distributed along a line (e.g., a charged

$fracdQdL$ (where dQ is a small charge over a small length dL)

- **Surface Charge Density ($sigma$):** Charge per unit area. Used for charges distributed over a surface (e.g., a charged plate).
  - Formula:
    $sigma =$
    $fracdQdA$ (where dQ is a small charge over a small area dA)
- **Volume Charge Density (rho):** Charge per unit volume. Used for charges distributed throughout a volume (e.g., a charged sphere).
  - Formula:
    $rho =$
    $fracdQdV$ (where dQ is a small charge over a small volume dV)

These densities allow us to treat a small, infinitesimally small portion of the continuous charge as a point charge, for which we already know how to calculate the electric field.

Electric Field Due to a Uniformly Charged Rod on its Axis

Problem Statement

Consider a rod of total length L with a uniform charge distribution. We need to find the electric field at a point P located on the axis of the rod. Point P is at a distance R from one end (B) of the rod.

Approach to Solving

Since the charge is distributed continuously along the rod, we cannot directly use the formula for a point charge. Instead, we employ the method of integration:

1. **Consider a Small Element:** We take an infinitesimally small slice of the rod. This slice is so small that it can be treated as a point charge.
2. **Define Variables:**
   - Let the small slice have a length dx.
   - The charge on this small slice is dq.
   - Let x be the distance of this small slice from point P.
3. **Express dq in terms of Linear Charge Density:**
   - The rod has a uniform linear charge density
     $lambda =$
     $frac$Total Charge (Q)Total Length (L).

4. **Calculate Electric Field due to the Small Element:**
   - The electric field dE produced by this small charge dq at point P is given by Coulomb's Law for a point charge: dE = $frackcdotdqx^2$
   - Substituting dq = $lambda$ $cdotdx$: dE = $frackcdotlambdacdotdxx^2$ where k is Coulomb's constant (k = $frac14piepsilon_0$).

Integration to Find Total Electric Field

To find the total electric field E at point P, we sum up the contributions from all such infinitesimally small slices along the entire length of the rod. This summation is performed using integration.

1. **Setting up the Integral:** E = $intdE$ = int $frackcdotlambdacdotdxx^2$
2. **Determining the Limits of Integration:**
   - The variable x represents the distance of a slice from point P.
   - When the slice is at the closer end B, its distance from P is R. So, the lower limit is x = R.
   - When the slice is at the farther end A, its distance from P is R + L. So, the upper limit is x = R + L.
   - Therefore, the integral becomes: E = k $lambda$ $int_R^{R+L}x^{-2}dx$
3. **Performing the Integration:**
   - The integral of x^{-2} with respect to x is $fracx^{-1}-1$ = − $frac1x$.

   E = k $lambda$ $left[-$ $frac1x$ $right]_R^{R+L}$
4. **Applying the Limits:** E = k $lambda$

$right) -$
$left(-$
$frac1R$
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$right] E = k$
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$frac1R -$
$frac1R + L$
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$lambda$
$left[$
$frac(R + L) - RR(R + L)$
$right] E = k$
$lambda$
$left[$
$fracLR(R + L)$
$right]$

5. **Final Expression in terms of Total Charge Q:**
   - We know that $lambda = fracQL$. Substituting this into the equation: $E = k left(fracQL right) left[fracLR(R + L) right]$
   - The L in the numerator and denominator cancels out: $E = frackQR(R + L)$

Summary of Result

The electric field on the axis of a uniformly charged rod at a point P located at a distance R from the closer end is:

$E = frackQR(R + L)$

Where:

- k is Coulomb's constant ($k = frac14piepsilon_0$)

- L is the total length of the rod.

This formula is crucial for understanding how electric fields are calculated for continuous charge distributions, emphasizing the role of integration in summing up infinitesimal contributions.

Electric Field due to a Charged Rod

Formula for Electric Field

The electric field (E) due to a charged rod at a point along its axis is given by the formula:

$$E = \frac{KQ}{R(L+R)}$$

Where:

- K is Coulomb's constant (9×10^9 Nm^2/C^2)
- Q is the total charge on the rod
- R is the distance from the closer end of the rod to the point
- L is the length of the rod

Relation to Point Charge Formula

This formula can be understood as a generalization of the electric field due to a point charge. The electric field due to a point charge is $E = \frac{KQ}{R^2}$.

If we consider a point very far away from the rod, such that the distance R is much greater than the length L ($R \gg L$), then:

- The term $(L + R)$ can be approximated as R (since L is negligible compared to R).
- Substituting this into the formula, we get $E \approx \frac{KQ}{R(R)} = \frac{KQ}{R^2}$.

This shows that from a large distance, a charged rod behaves like a point charge. This perspective helps in understanding the concept and memorizing the formula by linking it to known principles.

Example Problem

Given values:

- Electric field (E) = 10^9 N/C (Note: The transcript mentions a typing mistake and clarifies it should be Newton per Coulomb)
- Coulomb's constant (K) = 9×10^9 Nm^2/C^2
- Charge (Q) = 2 C
- Distance from closer end (R) = 3 m

Substitute the given values: $10^9 = \frac{(9\times10^9)\times2}{3(L+3)}$

Divide both sides by 10^9: $1 = \frac{9\times2}{3(L+3)}$

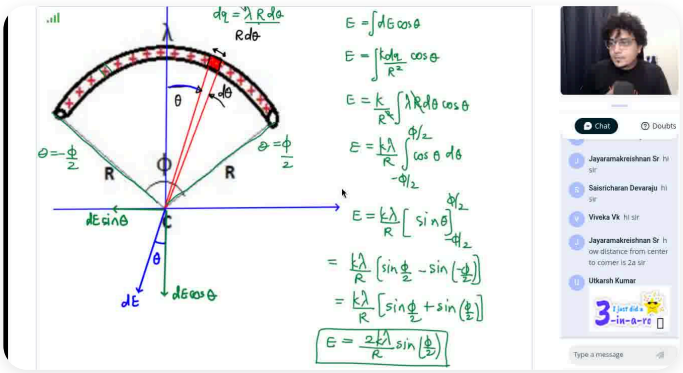
Simplify the right side: $1 = \frac{18}{3(L+3)}$ $1 = \frac{6}{L+3}$

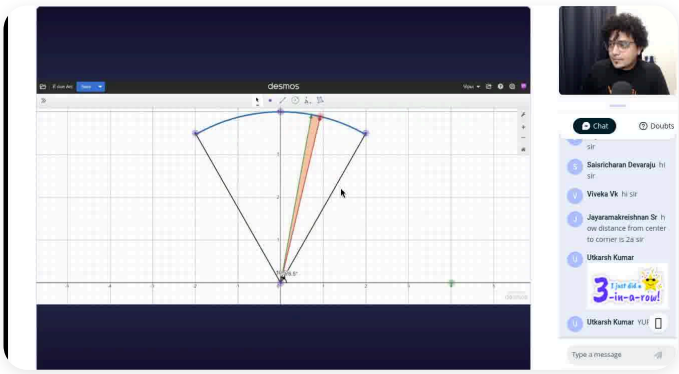
Multiply both sides by $(L + 3)$: $L + 3 = 6$

Subtract 3 from both sides: $L = 6 - 3$ $L = 3$ m

Therefore, the length of the rod is 3 m. This corresponds to option D.

Electric Field Due to a Uniformly Charged Arc





Formula for Electric Field at the Center of an Arc

For an arc with uniform linear charge density λ, radius R, and subtending an angle ϕ at its center, the electric field (E) produced at the center is given by: $E = \frac{2k\lambda}{R} \sin\left(\frac{\phi}{2}\right)$ Where:

- k is Coulomb's constant.
- λ is the linear charge density.
- R is the radius of the arc.
- ϕ is the total angle subtended by the arc.

Derivation of the Formula

To derive this formula, we consider a small element of the arc and integrate its contribution to the electric field.

1. **Consider a Tiny Slice:** We take a small element of the arc that subtends a tiny angle $d\theta$ at the center. The length of this tiny arc element is $ds = Rd\theta$.
2. **Charge on the Tiny Slice:** The charge dq on this tiny element is given by the linear charge density multiplied by its length: $dq = \lambda \cdot ds = \lambda R d\theta$

4. **Resolution of Electric Field Components:**
    - If we place the center of the arc at the origin and the bisector of the angle along the y-axis, a tiny element at an angle θ from the y-axis will produce an electric field dE directed away from it (if positive charge). This dE can be resolved into two components:
        - Horizontal component: $dE \sin\theta$
        - Vertical component: $dE \cos\theta$
5. **Cancellation of Horizontal Components:** Due to symmetry, for every element on one side of the bisector, there is a corresponding element on the other side. The horizontal components ($dE \sin\theta$) from these symmetric elements will be equal in magnitude and opposite in direction, thus cancelling each other out. Therefore, the net electric field will only be due to the sum of the vertical components.
6. **Integration of Vertical Components:** The net electric field E is the integral of the vertical components ($dE \cos\theta$) over the entire arc: $E = \int dE \cos\theta$ Substitute the expression for dE: $E = \int \frac{k\lambda}{R} \cos\theta d\theta$
7. **Setting Integration Limits:** If the total angle subtended by the arc is ϕ, and we consider the y-axis as the bisector, the angle θ will vary from $-\phi/2$ to $+\phi/2$. This ensures that we cover the entire arc symmetrically around the y-axis. $E = \frac{k\lambda}{R} \int_{-\phi/2}^{\phi/2} \cos\theta d\theta$
8. **Performing the Integration:** The integral of $\cos\theta$ with respect to θ is $\sin\theta$. $E = \frac{k\lambda}{R}[\sin\theta]_{-\phi/2}^{\phi/2}$ Apply the limits of integration: $E = \frac{k\lambda}{R}[\sin(\phi/2) - \sin(-\phi/2)]$ Since $\sin(-\theta) = -\sin(\theta)$: $E = \frac{k\lambda}{R}[\sin(\phi/2) - (-\sin(\phi/2))]$ $E = \frac{k\lambda}{R}[\sin(\phi/2) + \sin(\phi/2)]$ $E = \frac{k\lambda}{R}[2\sin(\phi/2)]$ $E = \frac{2k\lambda}{R} \sin\left(\frac{\phi}{2}\right)$

This confirms the formula for the electric field at the center of a uniformly charged arc.

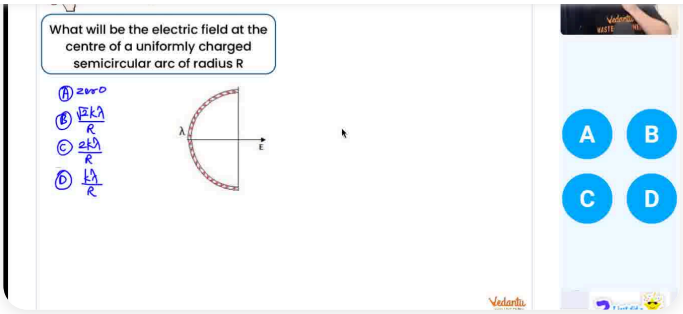
Direction of Electric Field for a Charged Arc

Direction for Positively Charged Arc

- If the arc is positively charged, the electric field at the center will be directed **away from the arc**.
- The direction will always bisect the angle ϕ subtended by the arc.

Direction for Negatively Charged Arc

- If the arc is negatively charged, the electric field at the center will be directed **towards the arc**.
- Similar to the positive case, the direction will still bisect the angle ϕ subtended by the arc.



Problem Statement

What will be the electric field at the centre of a uniformly charged semicircular arc of radius R?

(A) zero (B) $\frac{\sqrt{2}k\lambda}{R}$ (C) $\frac{2k\lambda}{R}$ (D) $\frac{k\lambda}{R}$

Solution

To find the electric field at the center of a uniformly charged semicircular arc, we use the general formula for the electric field due to a uniformly charged arc:

$$E = \frac{2k\lambda}{R} \sin\left(\frac{\phi}{2}\right)$$

Where:

- E is the electric field.
- k is Coulomb's constant ($k = \frac{1}{4\pi\epsilon_0}$).
- λ is the linear charge density of the arc.
- R is the radius of the arc.
- ϕ is the angle subtended by the arc at the center.

For a semicircular arc, the angle subtended at the center is 180° (or π radians). Therefore, $\phi = 180^\circ$.

Substitute the value of ϕ into the formula:

$$E = \frac{2k\lambda}{R} \sin\left(\frac{180^\circ}{2}\right)$$

$$E = \frac{2k\lambda}{R} \sin(90^\circ)$$

Since $\sin(90^\circ) = 1$, the equation simplifies to:

$$E = \frac{2k\lambda}{R}(1)$$

$$E = \frac{2k\lambda}{R}$$

a uniformly charged arc.

Homework and Important Formulas

Important Formulas

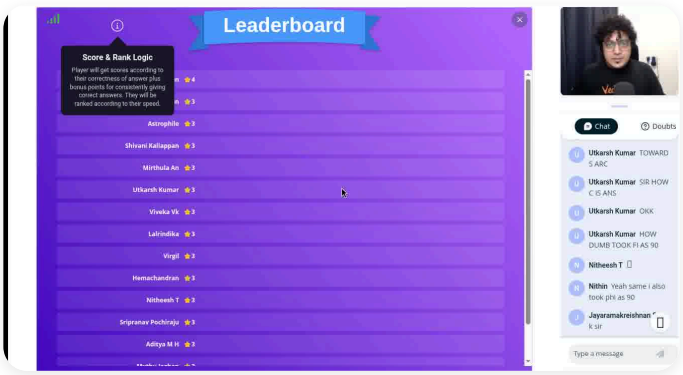
Before the next class, students are required to copy a list of important formulas into their notebooks. Some of these formulas have already been derived and discussed, while others will be covered in the upcoming session. This pre-copying ensures that class time can be dedicated to conceptual discussions and problem-solving rather than note-taking.

Homework Assignments

- **Assignment Number:** Question number 15.
- **Electrostatics 2 Homework:** Questions 16 to 21.
- **Extra Questions:** Additional questions provided in the session notes, including two pending from the previous class and one new question from the current class.
- **Work from EL2 and EL3:** Students should also complete work from these sections.

Tatwa will be provided after the next class, which will help in solving these problems. All homework is designed to be solvable with the current knowledge.

Leaderboard and Telegram Channel



Leaderboard

Score & Rank Logic

- Players receive scores based on the correctness of their answers.
- Bonus points are awarded for consistently providing correct answers.
- Ranking is determined by the speed of response.

Top Performers

- Astrophile: 3

- Utkarsh Kumar: 3
- Viveka Vk: 3
- Lalrindika: 3
- Virgil: 3
- Hemachandran: 3
- Nitheesh T: 3
- Sripranav Pochiraju: 3
- Aditya MH: 3

Telegram Channel

The Telegram channel link for the batch will be shared. Students are encouraged to join for updates and discussions. Feedback on the session is also encouraged.

Did you find these notes helpful?